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Notes for the course of "Fiber Optics"

NONLINEAR EFFECTS IN OPTICAL FIBERS

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Introduction

So far Maxwell equations have been studied under the approximation of a linear medium:

$$\nabla \times \bar{e} = -\frac{\partial \bar{b}}{\partial t} \qquad\qquad \nabla \times \bar{h} = \frac{\partial \bar{d}}{\partial t}$$

Where $\bar{b} = \mu_0 \bar{h}$ and

$$\bar{d} = \varepsilon\, \bar{e} = \varepsilon_0 \varepsilon_r\, \bar{e} = \varepsilon_0 \left(1 + \chi^{(1)}\right) \bar{e} = \varepsilon_0 \bar{e} + \bar{p}$$

Where $\chi^{(1)}$ is called **linear susceptibility** and

$$\bar{p} = \varepsilon_0 \chi^{(1)} \bar{e}$$

is the **linear medium polarization.**

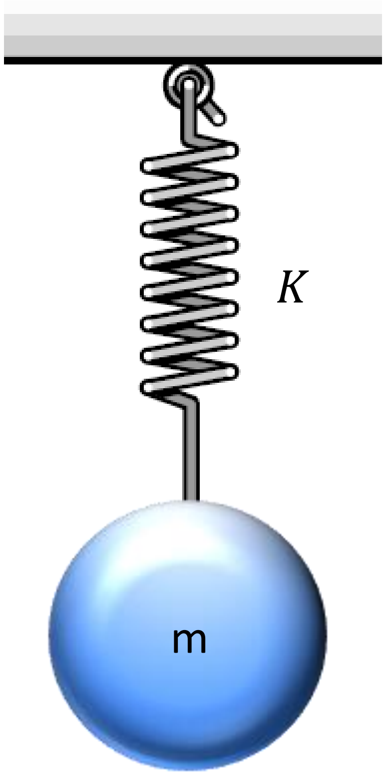
However, this is not always true and nonlinear terms should be accounted in some media when the intensity of the electric field becomes large enough.

Here, we first model nonlinear response of a medium with a simple approach, using a mechanical model of an atom excited by an electric field. The model, though not precise (quantum mechanics should be applied) is useful to understand the phenomenology associated to nonlinearity.

Non-conducting media – mechanical oscillator

In an isotropic **non-conducting** medium, such as dielectrics or semiconductors, the electrons are **bound** to the atoms. Like every other mass, electrons move under the action of forces. These forces are responsible for their movement as described in Newton's second law:

$$\boldsymbol{F} = m\boldsymbol{a}$$



The forces that an electron feels are:

- The Coulomb force: $-e\boldsymbol{E}$
- The Lorentz Force: $-e\boldsymbol{v} \times \boldsymbol{B}$
- The friction (damping) force: $-m\Gamma\boldsymbol{v}$
- The elastic force: $-K\boldsymbol{r}$

Therefore, we can write:

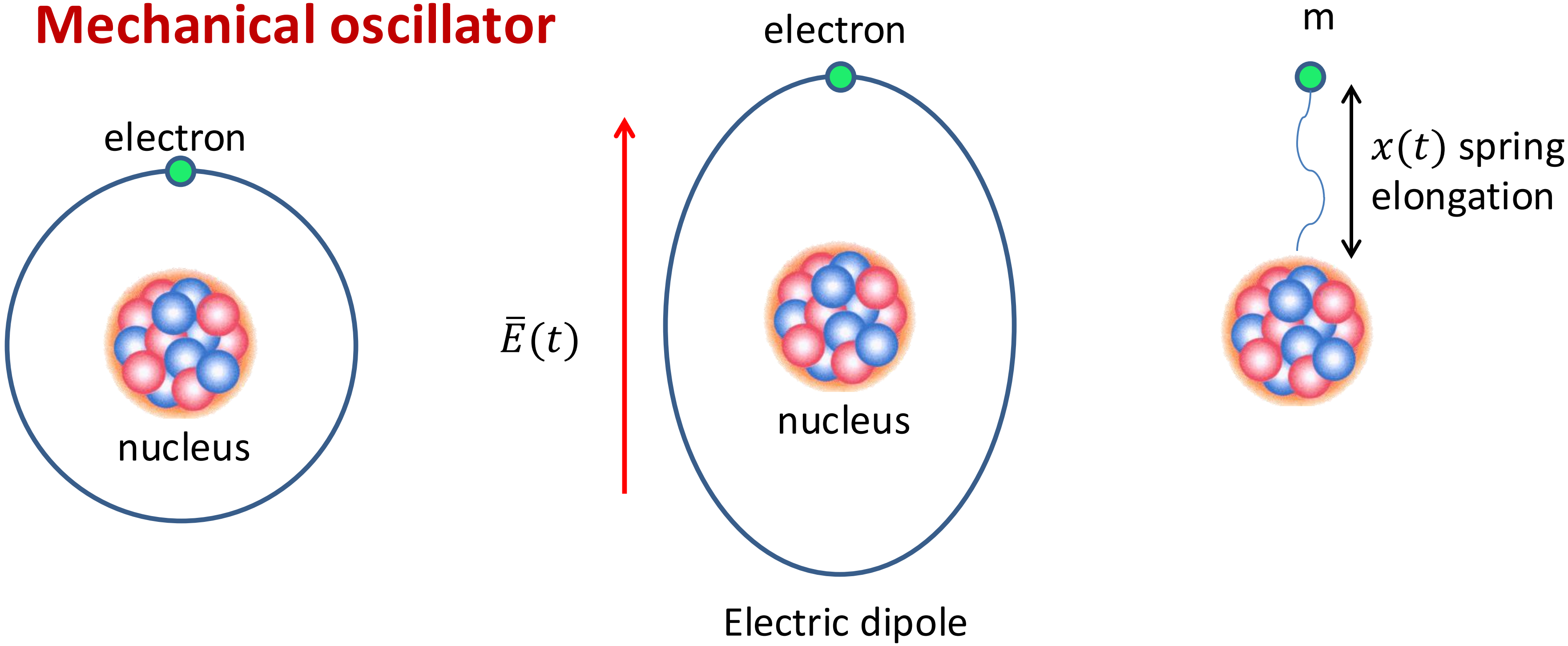
$$m\boldsymbol{a} = -e\boldsymbol{E} - e\boldsymbol{v} \times \boldsymbol{B} - K\boldsymbol{r} - m\Gamma\boldsymbol{v}$$

Hence, the motion of the electron can be properly described by the equation of a damped harmonic oscillator, which reads:

$$m\frac{d^2\boldsymbol{r}}{dt^2} + m\Gamma\frac{d\boldsymbol{r}}{dt} + K\boldsymbol{r} = -e\boldsymbol{E} - \cancel{e\boldsymbol{v} \times \boldsymbol{B}}$$

In the most common applications, the Lorentz force can be **neglected**.

Mechanical oscillator



Setting $K = m\Omega^2$, the harmonic equation of a charged mass forced by the electric field and bounded to the nucleus by a spring (modelling nucleus attraction force) is:

$$m\left[\frac{d^2x}{dt^2} + 2\Gamma\frac{dx}{dt} + \Omega^2 x\right] = -e\,\bar{E}(t)$$

Where Ω is the undamped angular frequency and Γ is the damping factor.

Mechanical oscillator

Assume that the electric field is sinusoidal:

$$E(t) = E_0 \cos \omega_0 t = \frac{E_0}{2}\left(e^{-j\omega_0 t} + e^{j\omega_0 t}\right)$$

By Fourier transforming the previous equation:

$$m(-\omega^2 X + j2\omega\Gamma X + \Omega^2 X) = -e\frac{E_0}{2}[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

solving for $X(\omega)$ and anti-transforming, the asymptotic solution reads:

$$x(t) = -\frac{eE_0}{2m}\,\frac{e^{-j\omega_0 t}}{\Omega^2 - \omega_0^2 - j2\Gamma\omega_0} + c.c.$$

Hereinafter let us consider only the first term (i.e., the analytic signal) omitting the second that is simply the complex conjugate.

Each atom becomes an electric oscillating dipole which emits an electromagnetic wave.

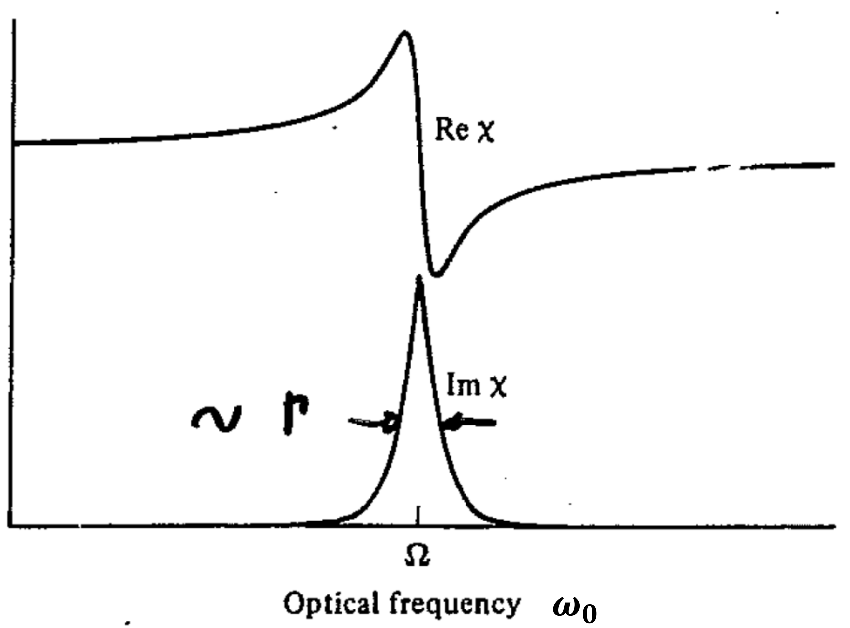
Mechanical oscillator

If N is the density of dipoles, the total medium polarization is

$$p(t) = -N\, e\, x(t) = \frac{N\, e^2\, E_0}{2m} \frac{e^{-j\omega_0 t}}{\Omega^2 - \omega_0^2 - j2\Gamma\omega_0} = \varepsilon_0 \chi^{(1)} E(t)$$

Where

$$\chi^{(1)}(\Omega, \omega_0) = \frac{N\, e^2}{\varepsilon_0\, m} \frac{1}{\Omega^2 - \omega_0^2 - j2\Gamma\omega_0}$$



Mechanical oscillator

The refractive index is

$$n = \sqrt{\epsilon} = \sqrt{1 + \chi^{(1)}}$$

Far from resonance, $Im\left[\chi^{(1)}\right] = 0$:

$$n = \sqrt{1 + Re[\chi^{(1)}]}$$

Close to the resonance, the refractive index becomes complex:

$$n = n' - jn''$$

The imaginary part represents gain/attenuation, in fact the propagation vector (of a plane wave) is given by:

$$\kappa = \frac{\omega n}{c_0} = \frac{\omega(n' - jn'')}{c_0} = \beta - j\alpha$$

and a propagating wave behaves like:

$$\exp(-j\kappa z) = \exp(-j\beta z - \alpha z)$$

Nonlinear mechanical oscillator

Away from resonance, we have $\omega_0 \neq \Omega$, hence $\left|\Omega^2 - \omega_0^2\right| \gg 2\Gamma\omega_0$ and the damping term can be neglected.

The response is not always linear with the elongation. Let consider:

$$m\left[\frac{d^2x}{dt^2} + \Omega^2 x\,(1 + \delta \cdot x)\right] = -e\,\bar{E}(t)$$

If δ is small, a perturbation method can be used, and the solution can be written as an expansion:

$$x(t) = x_0(t) + \delta \cdot x_1(t) + O(\delta^2)$$

Substituing into the equation and separating the contributions at different orders of the expansion:

$$O(\delta^0) \quad \rightarrow \quad \frac{d^2x_0}{dt^2} + \Omega^2 x_0 = -\frac{e}{m}E(t)$$

$$O(\delta^1) \quad \rightarrow \quad \frac{d^2x_1}{dt^2} + \Omega^2 x_1 = -\Omega^2 x_0^2(t)$$

The first contribution is the linear response. The second contribution is still the equation of a linear oscillator, but the forcing term is x_0^2 which **has harmonics at $2\omega_0$**.

Nonlinear mechanical oscillator

Suppose that

$$E(t) = E_1 \cos \omega_1 t + E_2 \cos \omega_2 t$$

Since we are far from resonance ($\left|\Omega^2 - \omega_{1,2}^2\right| \gg 2\Gamma\omega_{1,2}$), the solution at order δ^0 (no perturbation) reads:

$$x_0(t) = \frac{e}{m}\left(\frac{E_1 \cos \omega_1 t}{\Omega^2 - \omega_1^2} + \frac{E_2 \cos \omega_2 t}{\Omega^2 - \omega_2^2}\right)$$

The forcing term of the equation at order δ^1 then is:

$$-\Omega^2 x_0^2(t) = -\frac{\Omega^2 e}{m}\left(\frac{E_1 \cos \omega_1 t}{\Omega^2 - \omega_1^2} + \frac{E_2 \cos \omega_2 t}{\Omega^2 - \omega_2^2}\right)^2 =$$

$$-\frac{\Omega^2 e}{m}\left(\frac{E_1^2 \cos^2 \omega_1 t}{(\Omega^2 - \omega_1^2)^2} + \frac{E_2^2 \cos^2 \omega_2 t}{(\Omega^2 - \omega_2^2)^2} + 2\frac{E_1 E_2 \cos \omega_1 t \cos \omega_2 t}{(\Omega^2 - \omega_1^2)(\Omega^2 - \omega_2^2)}\right)$$

That contains sinusoidal waves at frequencies

$$0, \qquad 2\omega_1, \qquad 2\omega_2, \qquad \omega_1 + \omega_2, \qquad \omega_1 - \omega_2$$

Nonlinear mechanical oscillator

The solution at order δ^1 will contain terms oscillating at the same frequencies:

$$x_1(t) = X_0 + X_{2\omega_1}\cos(2\omega_1 t) + X_{2\omega_2}\cos(2\omega_2 t) + X_{\omega_1+\omega_2}\cos[(\omega_1+\omega_2)t] + X_{\omega_1-\omega_2}\cos[(\omega_1-\omega_2)t]$$

Therefore, the medium polarization $p(t) = -N\, e\, x(t)$ will contain terms oscillating at second harmonics, frequency sum and difference of all the input waves.

Nonlinear medium polarization

The simplified mechanical model shows that the medium response (polarization, its polarization) may be not linear. In general, this means that (neglecting dispersive effects)

$$\bar{p}(t) = \bar{p}\big(\bar{e}(t)\big)$$

In the limit of small nonlinearities, this expression can be expanded in Taylor series; for each component of the medium polarization, we can write ($i = 1, 2, 3$)

$$p_i(\bar{e}) = p_i(0) + \sum_{J=1}^{3} \frac{\partial p_i}{\partial e_j} e_j + \sum_{j,k=1}^{3} \frac{\partial^2 p_i}{\partial e_j \partial e_k} e_j e_k + \sum_{j,k,h=1}^{3} \frac{\partial^3 p_i}{\partial e_j \partial e_k \partial e_h} e_j e_k e_h$$

This expression can be written in a more compact form using tensor multiplication.

$$\bar{p} = \varepsilon_0\left(\chi^{(1)}\bar{e} + \chi^{(2)} : \bar{e}\bar{e} + \chi^{(3)} \vdots \bar{e}\bar{e}\bar{e} + \cdots\right)$$

Nonlinear medium polarization

The **susceptibilities** $\chi^{(n)}$:

1. have **units of** $[m/V]^{n-1}$

2. depend on all medium resonances and on all input frequencies

3. are complex

4. **are tensors** of 3^{n+1} elements in general; for the specific case of the electric field, they have 9, 18 and 27 elements, respectively.

Remember that the wave equation reads:

$$\nabla^2 \bar{e} - \frac{1}{c_0^2}\frac{\partial^2 \bar{e}}{\partial t^2} = \mu_0 \frac{\partial^2 \bar{p}}{\partial t^2}$$

so, it is very difficult to solve it when $\bar{p}$ is nonlinear. Simplifications will be introduced.

Nonlinear medium polarization

Summary of some relevant effects due to susceptibilities.

Linear term $\chi^{(1)}$

- in all media
- refractive index (and its dispersion), gain/attenuation

Quadratic term $\chi^{(2)}$

- in crystalline media
- second harmonic generation, sum and difference of frequencies, parametric amplification

Cubic term $\chi^{(3)}$

- in all media, including silica glass
- Third harmonic generation, self-and cross-phase modulation, parametric amplification and wavelength conversion, Raman scattering.

Nonlinear medium polarization

Quadratic nonlinearities are not present in media that are centro-symmetric, like **glasses**. A **centro-symmetric** medium satisfies the following condition:

> *If the electric field is sign-reversed, the medium polarization must also be sign-reversed because of the symmetry of the molecules.*

$$\bar{e} \to -\bar{e} \quad \Rightarrow \quad \bar{p} \to -\bar{p}$$

However:

$$\bar{p} = \varepsilon_0\left(\chi^{(1)}\bar{e} + \chi^{(2)}\colon \bar{e}\bar{e} + \chi^{(3)} \vdots \bar{e}\bar{e}\bar{e} + \cdots\right)$$

So

$$\bar{e} \to -\bar{e} \quad \Rightarrow \quad \bar{p} \to -\bar{p} \quad \Rightarrow \quad \chi^{(2n)} = 0$$

for all integers n.

Propagation in a cubic medium

We now proceed to the derivation of an equation to describe the propagation when linear and cubic nonlinear effects are present.

The cubic term gives rise to effects at two frequencies when the input field is made by only one frequency:

$$e(t) = Re[E_0 \exp(j\omega_0 t - j\beta z)] = \frac{1}{2}[E_0 \exp(j\omega_0 t - j\beta z) + \mathrm{c.c.}]$$

$$p(t) = \varepsilon_0 \chi^{(3)} \vdots \bar{e}\bar{e}\bar{e} = \frac{1}{8}\varepsilon_0 \chi^{(3)}[E_0 \exp(j\omega_0 t - j\beta z) + \mathrm{c.c.}]^3$$

The last formula introduces several simplifications:

1) The electric field is linearly polarized everywhere
2) The tensor product has been reduced to a scalar product (valid in isotropic media)

Then:

$$p(t) = \frac{1}{8}\varepsilon_0 \chi^{(3)}[E_0^3 \exp(j3\omega_0 t - j3\beta z) + 3|E_0|^2 E_0 \exp(j\omega_0 t - j\beta z) + \mathrm{c.c.}]$$

Propagation in a cubic medium

$$p(t) = \frac{1}{8}\varepsilon_0\chi^{(3)}[E_0^3 \exp(j3\omega_0 t - j3\beta z) + 3\mathrm{E}_0^2\mathrm{E}_0^* \exp(j\omega_0 t - j\beta z) + \mathrm{c.c.}]$$

The medium polarization has two kinds of terms

1. a term oscillating at the same input frequency. This term oscillates also at the same frequency of the linear polarization ($\chi^{(1)}$); therefore, it generates a field that is in phase with the forcing one

2. a **third harmonic** term oscillating at 3 times the input frequency. Notice that this term propagates according to the factor $\exp(j3\omega_0 t - j3\beta z)$, where $\beta = \beta(\omega_0)$ is the propagation constant at the input frequency. However, a wave at frequency $3\omega_0$ should have propagation constant $\beta(3\omega_0)$ and since in most cases we have $\beta(3\omega_0) \neq 3\beta(\omega_0)$, the polarization field at $3\omega_0$ would not be in phase with the corresponding electric field at the same frequency. This condition is called **phase mismatch.** In order words, we can also say that the third harmonic of the medium polarization is forcing an electromagnetic wave that cannot propagate in the medium. As a consequence, the third harmonic of the electromagnetic field is not efficiently generated and can be **neglected**.

Propagation in a cubic medium

Given that there are only fields at frequency ω_0, we can write Maxwell's eq. in the domain of complex phasors:

$$\nabla \times \bar{E} = -j\omega\bar{B} = -j\omega\mu_0\bar{H} \qquad \nabla \times \bar{H} = j\omega\bar{D}$$

where

$$\bar{D} = \varepsilon_0\left(1 + \chi^{(1)}\right)\bar{E} + \varepsilon_0\chi^{(3)} \vdots \bar{E}\bar{E}\bar{E} = \varepsilon_0\left(1 + \chi^{(1)} + \chi^{(3)}|\bar{E}|^2\right)\bar{E}$$

Evaluating $\nabla \times \nabla \times \bar{E}$ and using the properties of the ∇ operator, we reach the Helmholtz equation

$$\nabla^2\bar{E} + \frac{\omega^2}{c_0^2}\left(1 + \chi^{(1)}\right)\bar{E} = -\frac{\omega^2}{c_0^2}\chi^{(3)} \vdots \bar{E}\bar{E}\bar{E} = -\frac{3\omega^2}{4c_0^2}\chi^{(3)}|\bar{E}|^2\,\bar{E}$$

Where in the last equality we used the fact that the third harmonic can be neglected.

Propagation in a cubic medium

To solve the Helmholtz equation, a perturbation approach is used, into the effects of nonlinearity are considered so weak to not affect the mode structure but only the field envelope along the propagation direction z.

Let us assume the following solution:

$$\bar{E}(x,y,z) = F(x,y)\,\hat{A}(z)e^{-j\beta z}\hat{x}$$

Substituting into the Helmholtz equation

$$(\nabla_{\mathrm{t}}^2 F)\hat{A}\mathrm{e}^{-\mathrm{j}\beta z} + F\frac{\partial^2}{\partial z^2}\left(\hat{A}\mathrm{e}^{-\mathrm{j}\beta z}\right) + k^2 F\,\hat{A}\mathrm{e}^{-\mathrm{j}\beta z} = -\frac{3\omega^2}{4c_0^2}\chi^{(3)}\left|F\hat{A}\right|^2 F\hat{A}\mathrm{e}^{-\mathrm{j}\beta z}$$

where

$$k^2 = \omega^2\mu_0\varepsilon = \frac{\omega^2}{c_0^2}\left(1+\chi^{(1)}\right)$$

Propagation in a cubic medium

Explicitly calculating the second-order derivative and simplifying the propagation factor we find:

$$(\nabla_{\mathrm{t}}^2 F)\hat{A} + F\left(\frac{\partial^2 \hat{A}}{\partial z^2} - j2\beta\frac{\partial \hat{A}}{\partial z} - \beta^2 \hat{A}\right) + k^2 F\,\hat{A} = -\frac{3\omega^2}{4c_0^2}\chi^{(3)}\left|F\hat{A}\right|^2 F\hat{A}$$

that can be rearranged as

$$[\nabla_{\mathrm{t}}^2 F + (k^2 - \beta^2)F]\hat{A} + F\left(\frac{\partial^2 \hat{A}}{\partial z^2} - j2\beta\frac{\partial \hat{A}}{\partial z}\right) = -\frac{3\omega^2}{4c_0^2}\chi^{(3)}\left|F\hat{A}\right|^2 F\hat{A}$$

We recognize that in a linear fiber, the equation

$$\nabla_{\mathrm{t}}^2 F + (k^2 - \beta^2)F = 0$$

is the Helmholtz equation of the fiber modes; therefore, owing to the perturbation approach, we assume that the first term in square brackets is negligible also when there are weak nonlinear effects. For the same reason, we can also assume that $\hat{A}(z)$ is **slowly varying** along z with respect to the scale distance of the wavelength, which means:

$$\left|\frac{\partial^2 \hat{A}}{\partial z^2}\right| \ll \beta\left|\frac{\partial \hat{A}}{\partial z}\right|$$

Propagation in a cubic medium

In conclusion, we get the following equation:

$$F\frac{\partial \hat{A}}{\partial z} = -j\frac{3\omega^2}{8\beta c_0^2}\chi^{(3)}\left|F\hat{A}\right|^2 F\hat{A}$$

Multiplying by F^* on both sides and integrating in the transverse plane $\{x, y\}$ eliminates the dependence on the transverse coordinates:

$$\frac{\partial \hat{A}}{\partial z}\int_S |F|^2 dx\, dy = -j\left|\hat{A}\right|^2 \hat{A}\int_S |F|^4 dx\, dy$$

Which yields

$$\frac{\partial \hat{A}}{\partial z} = -j\left(\frac{3\omega^2\chi^{(3)}}{8\beta c_0^2}\,\frac{\int_S |F|^4 dx\, dy}{\int_S |F|^2 dx\, dy}\right)\left|\hat{A}\right|^2\hat{A}$$

About the physical units

Recalling that

$$\bar{E}(x,y,z) = F(x,y)\,\hat{A}(z)e^{-j\beta z}\hat{x} \qquad \longleftrightarrow \qquad [V/m]$$

we note that the physical units of F and $\hat{A}$ are arbitrary as long as the product $F\hat{A}$ has physical units of Volt per meter. It is however quite customary to write the propagation equation for a quantity whose squared modulus is the optical power flowing in the fiber. Given the Poynting vector, this power reads

$$\bar{P} \approx \frac{\beta}{2\omega\mu_0}|\bar{E}|^2\hat{z} \qquad \Rightarrow \qquad P(z) = \int_S (\bar{P}\cdot\hat{z})dx\,dy = \frac{\beta}{2\omega\mu_0}\left|\hat{A}(z)\right|^2 \int_S |F|^2 dx\,dy$$

where $P(z)$ is the **power** flowing in the fiber. Accordingly, we introduce the variable

$$\hat{p}(z) = \hat{A}(z)\left(\frac{\beta}{2\omega\mu_0}\int_S |F|^2 dx\,dy\right)^{1/2}$$

for which we clearly have $|\hat{p}(z)|^2 = P(z)$, as sought.

Nonlinear coefficient

Taking the z-derivative of $\hat{\mathrm{p}}(z)$, we reach the differential equation

$$\boxed{\frac{\partial \hat{p}}{\partial z} = -j\gamma|\hat{p}|^2\hat{p}}$$

where we introduced the **nonlinear coefficient ($W^{-1}m^{-1}$)**

$$\gamma = \frac{3\omega^3\mu_0\chi^{(3)}}{4\beta^2c_0^2}\,\frac{\int_S |F|^4 dx\,dy}{\left(\int_S |F|^2 dx\,dy\right)^2}$$

The quantity

$$A_{eff} = \frac{\left(\int_S |F|^2 dx\,dy\right)^2}{\int_S |F|^4 dx\,dy}$$

has physical units of m^2 and is therefore called **effective area**; indeed, this quantity provides a measure of how much the mode is spread across the fiber section.

Nonlinear refractive index

Moreover, remember that

$$\nabla^2 \bar{E} + \frac{\omega^2}{c_0}\left(1 + \chi^{(1)} + \frac{3}{4}\chi^{(3)}|\bar{E}|^2\right)\bar{E} = 0$$

which means that the squared refractive index reads

$$n^2 = \varepsilon_r = 1 + \chi^{(1)} + \frac{3}{4}\chi^{(3)}|\bar{E}|^2$$

Hence,

$$n = \sqrt{\varepsilon_r} = \sqrt{1 + \chi^{(1)}}\left(1 + \frac{3\chi^{(3)}|\bar{E}|^2}{4\left(1 + \chi^{(1)}\right)}\right)^{1/2} \approx \sqrt{1 + \chi^{(1)}} + \frac{3\chi^{(3)}}{8\sqrt{1 + \chi^{(1)}}}|\bar{E}|^2$$

We want to express the refractive index n in function of the field intensity I, which is by definition the modulus of the Poynting vector:

$$I = |\bar{P}| = \frac{\beta}{2\omega\mu_0}|\bar{E}|^2$$

Nonlinear refractive index

so we have:

$$\boxed{n(I) \approx n_0 + n_2 I}$$

where

- $n_0 = \sqrt{1 + \chi^{(1)}}$ is the linear refractive index
- $n_2 = \left(3\omega\mu_0\chi^{(3)}\right)/(4\, n_0\beta)$ is the **nonlinear refractive index** which has units of $m^2\, W^{-1}$

Noticing that

$$\beta = \frac{\omega}{c_0} n_{eff} \approx \frac{\omega}{c_0} n_0$$

we can express the nonlinear coefficient as

$$\boxed{\gamma = \frac{\omega\, n_2}{c_0\, A_{eff}} \quad \leftrightarrow \quad [W^{-1} m^{-1}]}$$

Typical values

In single-mode optical fiber the typical value of n_2 is

$$n_2 \approx 2.6 \times 10^{-20}\ \ m^2/W$$

with small variation depending on the fiber type.

Moreover, **at 1550 nm** the effective area is

$$A_{eff} \approx 20\ - 100\,\mu m^2$$

Thus, the nonlinear coefficients is about

$$\boldsymbol{\gamma \approx 1 - 10\ \ W^{-1} km^{-1}}$$

Pulse propagation in a cubic medium

Recalling that

$$\bar{E}(x,y,z) = F'(x,y)\,\hat{p}(z)e^{-j\beta z}\hat{x} = \hat{a}(z)F'(x,y)\hat{x}$$

The actual complex amplitude of the wave is

$$\hat{a}(z) = \hat{p}(z)e^{-j\beta z}$$

and the corresponding differential equation can be easily found evaluating the derivative of $\hat{a}(z)$ and using the differential equation for $\hat{p}(z)$; the result reads

$$\frac{\partial \hat{a}}{\partial z} = -j\beta\hat{a} - j\gamma|\hat{a}|^2\hat{a}$$

where we used the fact that $|\hat{a}|^2 = |\hat{p}|^2$.

The above equation has been determined for a specific angular frequency ω; therefore, we know that it is also representing the spectrum of a wide-band signal:

$$\frac{\partial \hat{a}(\omega)}{\partial z} = -j\beta(\omega)\hat{a}(\omega) - j\gamma(\omega)|\hat{a}(\omega)|^2\hat{a}(\omega)$$

Pulse propagation in a cubic medium

$$\frac{\partial \hat{a}(\omega)}{\partial z} = -j\beta(\omega)\hat{a}(\omega) - j\gamma(\omega)|\hat{a}(\omega)|^2\hat{a}(\omega)$$

The propagation constant can be expressed in its Taylor series around the carrier frequency to highlight the **delay of propagation** and the **chromatic dispersion**:

$$\frac{\partial \hat{a}(\omega)}{\partial z} = -j\left(\beta_0 + \beta_1\omega + \frac{1}{2}\beta_2\omega^2\right)\hat{a}(\omega) - j\gamma|\hat{a}(\omega)|^2\hat{a}(\omega)$$

The evolution of the pulse in the time domain can now be obtained by anti-transforming the above equation; the nonlinear term, however, makes the task not trivial.

It can be shown that **by neglecting**

1. the frequency dependence of γ
2. higher order nonlinear effects related to the term $|\hat{a}(\omega)|^2\hat{a}(\omega)$

we get …

Pulse propagation in a cubic medium

$$\boxed{\frac{\partial a}{\partial z} = -j\beta_0 a - \beta_1 \frac{\partial a}{\partial t} + j\frac{1}{2}\beta_2 \frac{\partial^2 a}{\partial t^2} - j\gamma|a|^2 a}$$

which is known as the **nonlinear Schrödinger equation (NLSE)**.

In this equation we recognize that

- $\beta_1 = \partial\beta/\partial\omega = 1/v_g$ is the inverse of the group velocity
- $\beta_2 = \partial^2\beta/\partial\omega^2$ is the chromatic dispersion coefficient

More commonly, the immaterial phase term related to β_0 is included in $a(t)$, so that the NLSE reads

$$\boxed{\frac{\partial a}{\partial z} = -\beta_1 \frac{\partial a}{\partial t} + j\frac{1}{2}\beta_2 \frac{\partial^2 a}{\partial t^2} - j\gamma|a|^2 a}$$

Pulse propagation in cubic medium

The term proportional to β_1 and relative to the delay of propagation can be removed from the equation by introducing the following transformation

$$\tau = t - \beta_1 z \qquad\qquad \zeta = \mathrm{z}$$

We have:

$$\frac{\partial}{\partial t} = \frac{\partial}{\partial \tau} \qquad\qquad \frac{\partial}{\partial z} = \frac{\partial \tau}{\partial z}\frac{\partial}{\partial \tau} + \frac{\partial \zeta}{\partial z}\frac{\partial}{\partial \zeta} = \frac{\partial}{\partial \zeta} - \beta_1 \frac{\partial}{\partial \tau}$$

so that the NLSE becomes (we use again t and z for time and space coordinates)

$$\boxed{\frac{\partial a}{\partial z} = j\frac{1}{2}\beta_2 \frac{\partial^2 a}{\partial t^2} - j\gamma |a|^2 a}$$

Note that the above transformation corresponds to moving the reference frame along z at the speed $1/\beta_1$, so to “follow” the pulse.

Pulse propagation in cubic medium

Let us consider the effect of the **nonlinearity alone**, assuming a negligible dispersion (e.g. an optical fibers close to the zero-dispersion wavelength)

$$\frac{\partial a}{\partial z} = -j\gamma|a|^2 a$$

Notice that

$$\frac{\partial |a|^2}{\partial z} = \frac{\partial a}{\partial z}a^* + a\frac{\partial a^*}{\partial z} = -j\gamma|a|^4 + j\gamma|a|^4 = 0$$

Therefore, the nonlinear term does not modify the amplitude of the wave, but only the phase. This effect is called **self-phase modulation**.

The general solution can be found noticing that

$$\frac{\partial |a|^2}{\partial z} = 0 \quad \Rightarrow \quad |a(z,t)|^2 = |a(0,t)|^2 = P(t)$$

where $P(t)$ is the input pulse power.

Pulse propagation in cubic medium

Therefore, the NLSE can be written as

$$\frac{\partial a(z,t)}{\partial z} = -j\gamma|a(z,t)|^2 a(z,t) = -j\gamma P(t) a(z,t)$$

For each time t, this equation is linear in z and can be readily solved as

$$\boxed{a(z,t) = a(0,t)\,\exp[-\mathrm{j}\gamma P(t)\,z]}$$

confirming that the nonlinearity affects only the phase of the wave, originating a time- and power-dependent nonlinear phase

$$\Phi_{NL}(z,t) = -\gamma P(t)\,z$$

The effects of $\chi^{(3)}$ nonlinearity can then be described with the **nonlinear length**

$$L_{NL} = \frac{1}{\gamma P_0}$$

where P_0 is the peak power of the signal.

Pulse chirping in cubic medium

The time-dependence of the nonlinear phase

$$\Phi_{NL}(z,t) = -\gamma P(t)\, z$$

is associate also to a variation of the instantaneous frequency as

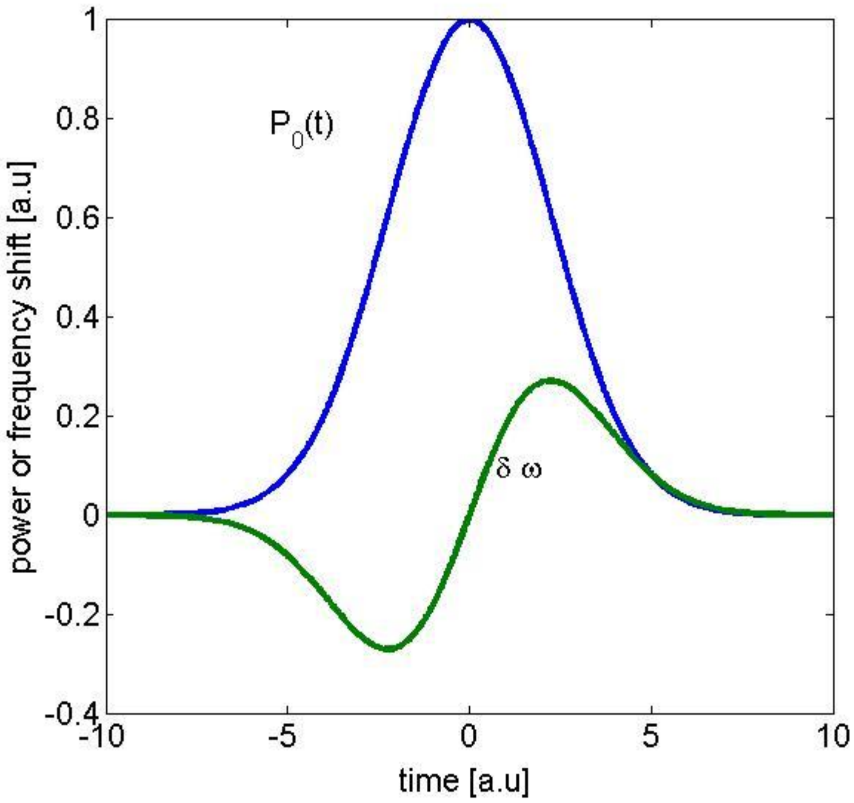
$$f(z,t) = \frac{1}{2\pi}\frac{\partial \Phi_{NL}}{\partial \tau} = -\frac{\gamma}{2\pi}\frac{dP}{d\tau}z$$

this shows that the nonlinearity chirps the pulse and that this **chirp grows linearly** with the distance of propagation.

Pulse propagation in cubic medium

As an example, let us assume a **Gaussian pulse**

$$a(0,t) = a_0 \exp\left(-\frac{t^2}{2T_0^2}\right) \quad \Rightarrow \quad P(t) = |a_0|^2 \exp\left(-\frac{t^2}{T_0^2}\right)$$



Notice that, while the pulse envelope does not change in time, the pulse phase does changes.

As a consequence, the spectrum of the broadens.

Moreover, the above effects increases linearly with the distance of propagation

Pulse propagation in cubic medium

For the Gaussian pulse, the maximum and minimum instantaneous frequency—i.e. the maximum and minimum of dP/dt, occurs for

$$t = \pm\frac{T_0}{\sqrt{2}} \quad \Rightarrow \quad \frac{dP}{dt} = \pm\sqrt{\frac{2}{e}}\,\frac{|\mathrm{a}_0|^2}{\mathrm{T}_0} \approx \pm 0.86\ \frac{|\mathrm{a}_0|^2}{\mathrm{T}_0}$$

Assuming:

- $\gamma = 10\ W^{-1}km^{-1}$
- $|a_0|^2 = P_0 = 100\ mW$
- $T = 100\ ps$ (pulse duration for an NRZ at 10 Gbit/s)

the maximum frequency variation is

$$f_{max} - f_{min} \approx 2.7\ GHz/km$$

and the nonlinear length is

$$L_{NL} \approx 1\ km$$

The effect of fiber attenuation

Attenuation can be easily included in the NLSE as

$$\frac{\partial a}{\partial z} = -\frac{\alpha}{2} a - \beta_1 \frac{\partial a}{\partial t} + j\frac{1}{2}\beta_2 \frac{\partial^2 a}{\partial t^2} - j\gamma |a|^2 a$$

where α describes power losses (hence the factor $1/2$) and $\alpha \approx \alpha_{dB}/4.34$.

Neglecting again the delay (β_1) and the chromatic dispersion (β_2), and considering the normalized amplitude

$$u(z,t) = \exp\left(\frac{\alpha}{2} z\right) a(z,t)$$

the equation become

$$\frac{\partial u}{\partial z} = -j\gamma \exp(-\alpha z)\, |u|^2 u$$

The effect of fiber attenuation

Again, we have that $d|u|^2/dz = 0$, so proceeding as before, we obtain

$$a(z,t) = a(0,t)e^{-(\alpha/2)z}\exp[j\Phi_{NL}(z,t)]$$

where now

$$\boxed{\Phi_{NL}(z,t) = -\gamma P(t)\frac{1-e^{-\alpha z}}{\alpha}}$$

The last factor shows that, because of fiber losses, the accumulated nonlinear phase is less than in the lossless case. Given a fiber of length L, we define the **effective length**

$$L_{eff} = \frac{1-e^{-\alpha L}}{\alpha}$$

as the length that a lossless fiber should have for the signal to accrue the same nonlinear phase it would accrue in the lossy fiber. Note that for $\alpha L \gg 1$, $L_{eff} \approx 1/\alpha$.

Recalling that $\alpha_{dB} \approx 4.34\alpha$, for a typical attenuation of $0.2\ dB/km$, as the fiber length increases, the effective length L_{eff} varies from L to the asymptotic values of about $22\ km$.

Chromatic dispersion

For comparison, we recall that chromatic dispersion (and all linear effects, in general) has a dual behavior, as it leaves the amplitude of the signal spectrum unchanged, while changing its phase:

$$\frac{\partial \hat{a}(\omega)}{\partial z} = -j\beta(\omega)\hat{a}(\omega) \quad \Rightarrow \quad \hat{a}(z,\omega) = \hat{a}(0,\omega)\exp[-j\beta(\omega)z]$$

For the same Gaussian pulse as before:

$$a(z,t) = \frac{a_0}{\sqrt{1 + js_{\beta 2}\frac{z}{L_D}}}\exp\left(j\frac{t^2/(2T_0^2)}{s_{\beta 2}\frac{z}{L_D}\left(1+\frac{L_D^2}{z^2}\right)}\right)\exp\left(-\frac{t^2/(2T_0^2)}{1+\frac{z^2}{L_D^2}}\right)$$

where

- $L_D = T_0^2/|\beta_2|$ is the **dispersion length**
- $s_{\beta 2} = \mathrm{sgn}(\beta_2)$

Solitons

The chromatic dispersion broadens the signal in time, while the Kerr nonlinearity broadens the signal spectrum. Both are distortions; however, they can self-compensate each other.

Consider the following normalization:

$$\xi = \frac{z}{L_D}, \qquad \tau = \frac{t}{T_0}, \qquad u = a\sqrt{\gamma L_D}$$

we have

$$\frac{\partial}{\partial t} = \frac{\partial \tau}{\partial t}\frac{\partial}{\partial \tau} = \frac{1}{T_0}\frac{\partial}{\partial \tau} \quad \Rightarrow \quad \frac{\partial^2}{\partial t^2} = \frac{1}{T_0^2}\frac{\partial^2}{\partial \tau^2} \qquad\qquad \frac{\partial}{\partial z} = \frac{\partial \xi}{\partial z}\frac{\partial}{\partial \xi} = \frac{1}{L_D}\frac{\partial}{\partial \xi}$$

and the NLSE becames:

$$\frac{\partial u}{\partial \xi} = j\,\mathrm{sgn}(\beta_2)\frac{1}{2}\frac{\partial^2 u}{\partial \tau^2} - j|u|^2 u$$

Solitons

When the fiber has **anomalous dispersion—i.e. $\boldsymbol{\beta_2 < 0}$**, the equation

$$\frac{\partial u}{\partial \xi} = -j\,\frac{1}{2}\frac{\partial^2 u}{\partial \tau^2} - j|u|^2 u$$

has a closed-form solution

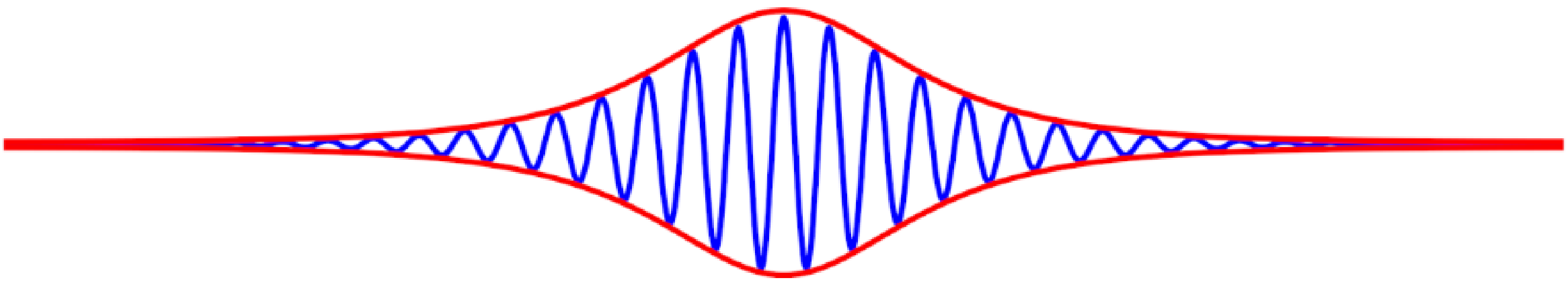
$$u(\xi,\tau) = u_0\,\mathrm{sech}(u_0\,\tau)\,\exp\left(-ju_0^2\,\frac{\xi}{2}\right)$$

which corresponds to ($P_0 = a_0^2 = \gamma L_D u_0^2$)

$$\boxed{a(z,t) = \sqrt{P_0}\,\mathrm{sech}\left(\sqrt{\frac{\gamma P_0}{|\beta_2|}}\cdot t\right)\exp\left(-j\,\frac{\gamma P_0}{2}\,z\right)}$$

This wave is called **soliton**. It propagates without distortion because the amplitude in time does not depend on z and there is no frequency modulation but only a phase shift (time independent but power dependent).

Solitons



$$a(z,t) = \sqrt{P_0}\,\mathrm{sech}\left(\sqrt{\frac{\gamma P_0}{|\beta_2|}}\cdot t\right)\exp\left(-j\frac{\gamma P_0}{2}z\right)$$

The soliton shape does not change during propagation. Similarly, the phase changes along the fiber but not with respect to time; this means that the pulse is not chirped and hence also the pulse spectrum doesn't vary during propagation.

Solitons

Solitons were thought to be a solution for long haul transmission (nonlinearity and dispersion self compensate). However, because of attenuation, the ideal soliton does not exist in reality.

Still, the interplay between nonlinearity and chromatic dispersion can be managed to attain a periodic (along the link) self-compensation of the dispersion. This approach is called **dispersion management** and requires the installation of fiber links along which the nonlinearity and the chromatic dispersion vary according to a specific design. This poses practical difficulties, owing to which dispersion management is used only in some undersea links and in only one terrestrial system (which however crosses the Australian desert, an environment not much different from the ocean floor).

Solitons might become very important for optical processing, as they are now observed in short waveguides.

Modulation instability

Neglecting losses, the NLSE reads:

$$\frac{\partial a}{\partial z} = j\frac{1}{2}\beta_2\frac{\partial^2 a}{\partial t^2} - j\gamma|a|^2 a$$

It is easy to prove that the CW wave

$$a(z,t) = a_{CW}(z) = \sqrt{\mathrm{P_0}}\exp(-\mathrm{j}\gamma P_0 z)$$

is a solution of the NLSE (note that, being CW, this solution is not affected by chromatic dispersion). It is interesting, to verify how stable this solution is with respect to perturbation; therefore, let consider the following wave

$$a(z,t) = \left[\sqrt{\mathrm{P_0}} + \eta(z,t)\right]\exp(-\mathrm{j}\gamma P_0 z)$$

where $\eta(z,t)$ is a small perturbation ($|\eta|^2 \ll P_0$). Inserting this expression in the NLSE, recalling that $a_{CW}(z)$ is a solution and neglecting terms proportional to orders of $|\eta|$ larger that 1 (i.e. neglecting higher-order perturbation), we find

$$\frac{\partial \eta}{\partial z} \approx j\frac{1}{2}\beta_2\frac{\partial^2 \eta}{\partial t^2} - j\gamma P_0(\eta + \eta^*)$$

Modulation instability

The previous equation is linear in η and can be Fourier-transformed to get (we omit the z-dependence for brevity)

$$\frac{\partial \hat{\eta}(\omega)}{\partial z} = -j\frac{1}{2}\beta_2\omega^2\hat{\eta}(\omega) - j\gamma P_0[\hat{\eta}(\omega) + \hat{\eta}^*(-\omega)]$$

The presence of the term $\hat{\eta}^*(-\omega)$ makes the analysis not straightforward. So let

$$\hat{\eta}(\omega) = \begin{cases} \hat{\eta}_+(\omega), & \omega \geq 0 \\ \hat{\eta}_-(\omega), & \omega < 0 \end{cases} \quad \Rightarrow \quad \hat{\eta}^*(-\omega) = \begin{cases} \hat{\eta}_-^*(\omega), & \omega > 0 \\ \hat{\eta}_+^*(\omega), & \omega \geq 0 \end{cases}$$

with this decomposition the differential equation for $\hat{\eta}(z,\omega)$ can be rearranged as

$$\frac{\partial}{\partial z}\begin{pmatrix} \hat{\eta}_+ \\ \hat{\eta}_-^* \end{pmatrix} = -j\begin{pmatrix} \frac{1}{2}\beta_2\omega^2 + \gamma P_0 & \gamma P_0 \\ -\gamma P_0 & -\frac{1}{2}\beta_2\omega^2 - \gamma P_0 \end{pmatrix}\begin{pmatrix} \hat{\eta}_+ \\ \hat{\eta}_-^* \end{pmatrix}$$

the general solution of this equation is a linear combination of terms proportional to $\exp(\pm v\, z)$, where $\pm v$ are the two eigenvalues of the matrix, with

$$v = \pm j\frac{1}{2}|\omega\beta_2|\sqrt{\omega^2 + \mathrm{sgn}(\beta_2)\omega_C^2}\,, \qquad \text{where} \quad \omega_C^2 = \frac{4\gamma P_0}{|\beta_2|}$$

Modulation instability

There are three cases:

$$v = \pm j \frac{1}{2} |\omega\beta_2| \sqrt{\omega^2 + \mathrm{sgn}(\beta_2)\omega_C^2}\,, \qquad \text{where} \quad \omega_C^2 = \frac{4\gamma P_0}{|\beta_2|}$$

Normal dispersion ($\boldsymbol{\beta_2 > 0}$)

The eigenvalues are always imaginary; as a consequence, the phase of the perturbation $\hat{\eta}(z, \omega)$ varies, whereas its amplitude does not change. This means that the perturbation remains small and hence the **solution is stable**.

Anomalous dispersion ($\boldsymbol{\beta_2 < 0}$) and $\boldsymbol{\omega > \omega_C}$

The eigenvalues are again imaginary, hence the **solution is stable**.

Anomalous dispersion ($\boldsymbol{\beta_2 < 0}$) and $\boldsymbol{\omega < \omega_C}$

The eigenvalues are real; the negative one describes a decaying trend of the perturbation, but the positive ones indicates that the perturbation grows exponentially along z; therefore the **solution is unstable**.

Modulation instability

Anomalous dispersion ($\boldsymbol{\beta_2 < 0}$) and $\boldsymbol{\omega < \omega_C}$

The positive eigenvalue describes a nonlinear gain equal to

$$g(\omega) = |\omega\beta_2|\sqrt{\omega_C^2 - \omega^2}\,, \qquad \text{where} \quad \omega_C^2 = \frac{4\gamma P_0}{|\beta_2|}$$

The maximum gain occurs for

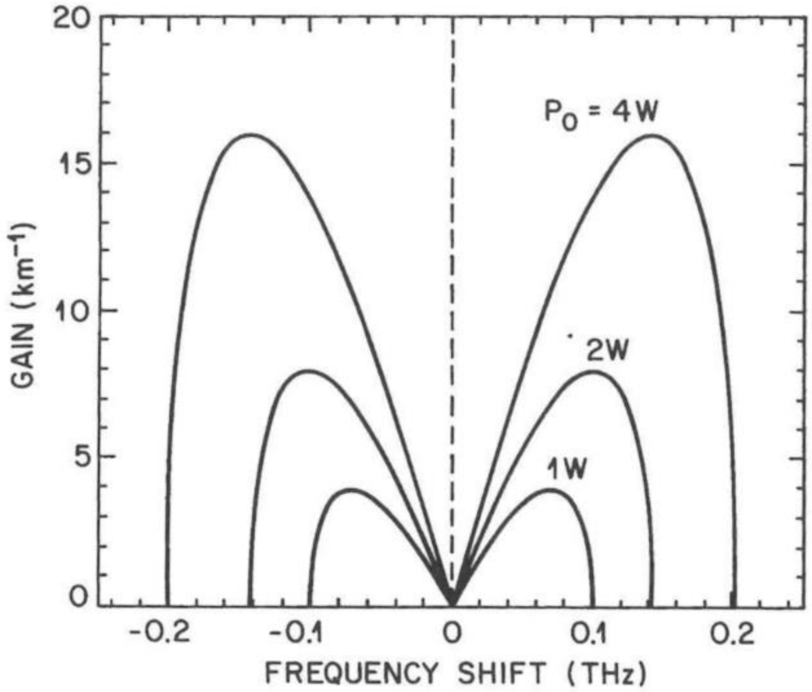
$$\omega_{max} = \pm\frac{\omega_c}{\sqrt{2}} = \pm\left(\frac{2\gamma P_0}{|\beta_2|}\right)^{1/2}$$

with a peak value of

$$g_{max} = \frac{1}{2}|\beta_2|\omega_C^2 = 2\gamma P_0$$

As a consequence of this gain, any small perturbation superposed to the ideal solution—e.g. amplifier, laser or quantum noise, is amplified making the wave unstable.

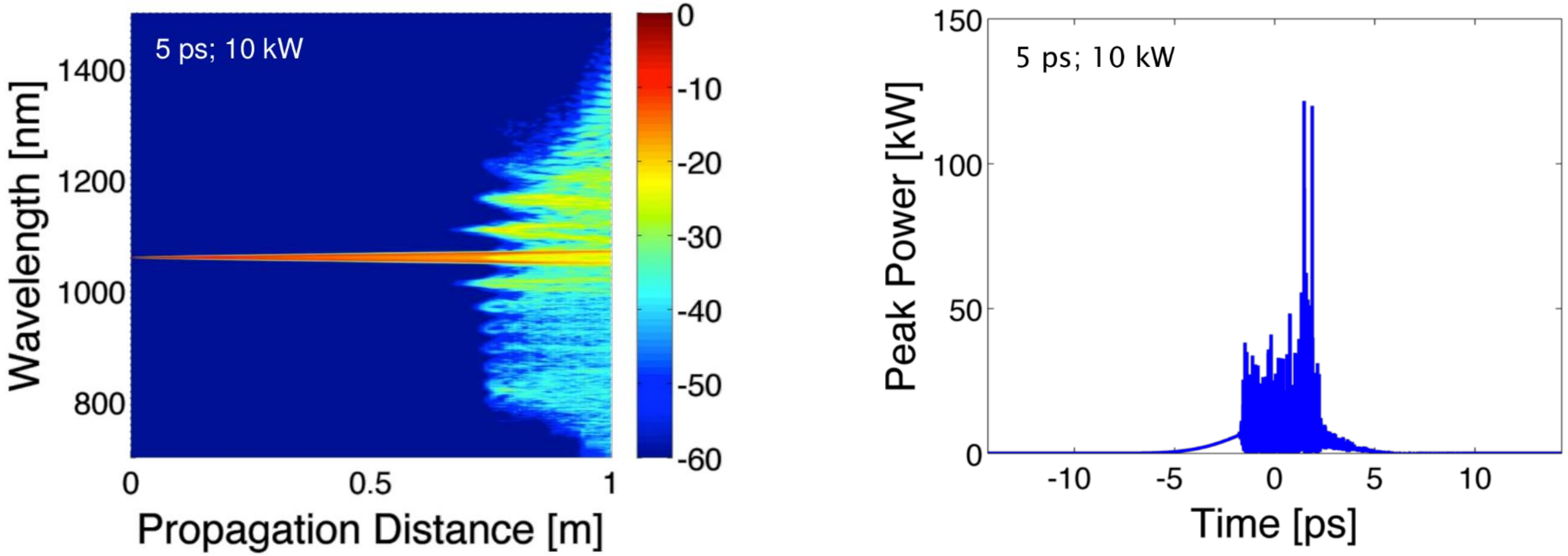
Modulation instability



Gain spectra of MI at different power level for an optical fiber with $\beta_2 = -20\ ps^2/km$ and $\gamma = 2\ W^{-1}/km$.

[source: Agrawal, "Nonlinear fiber optics"]

Modulation instability



Effects of MI on a short high-power pulse: amplification of quantum noise.

[credits: Alexander Heidt]

Modulation instability

Modulation instability affect the performance of long-haul communication systems, especially for NRZ ones.

Phenomena that are similar to modulation instability can take place whenever there is a periodicity along the fiber link. This could be the case, for example, of period amplifiers or periodic compensation of the chromatic dispersion. These periodicities can seed a modulation instability also for normal dispersion.

Cross-phase modulation (XPM)

So far, we have considered the effects of cubic nonlinearity on a single wave; it is interesting to analyze what happens when 2 (or more) waves at different frequencies interact.

When one of these waves is powerful enough to induce nonlinearity, the nonlinear refractive index variation will be experienced also by the other waves, sharing the same propagation medium. As a result, waves are coupled by means of nonlinearity.

The nonlinear medium polarization for isotropic cubic nonlinearity is

$$p_{NL}(t) = \varepsilon_0 \chi^{(3)} \vdots \bar{e}\bar{e}\bar{e}$$

Assuming that the field is the superposition of two waves (note that here $\beta_k = \beta(\omega_k)$)

$$e(t) = \frac{1}{2}\left[E_1 e^{j(\omega_1 t - \beta_1 z)} + E_2 e^{j(\omega_2 t - \beta_2 z)} + \mathrm{c.\,c.}\right]$$

the nonlinear medium polarization reads

$$p_{NL}(t) = \frac{1}{8}\varepsilon_0 \chi^{(3)}\left[E_1 e^{j(\omega_1 t - \beta_1 z)} + E_2 e^{j(\omega_2 t - \beta_2 z)} + \mathrm{c.\,c.}\right]^3$$

Cross-phase modulation (XPM)

Explicit calculation yields terms with the frequencies and propagation constants listed in the table.

#	Frequency	Prop. const.
1	$3\omega_1$	$3\beta_1$
2	$3\omega_2$	$3\beta_2$
3	$2\omega_1 + \omega_2$	$2\beta_1 + \beta_2$
4	$2\omega_2 + \omega_1$	$2\beta_2 + \beta_1$
5	$2\omega_1 - \omega_2$	$2\beta_1 - \beta_2$
6	$2\omega_2 - \omega_1$	$2\beta_2 - \beta_1$
7	ω_1	β_1
8	ω_2	β_2

By the same argument given when we considered a single frequency, we can say that the first 4 terms are (almost) always phase mismatched, so they can be neglected.

The fifth and sixth terms may suffer a similar problem; however, for a specific dispersion profile, there might be frequencies at which the waves are phase matched. In the special case in which this occurs, there is an effective creation of waves at two other frequencies. This phenomenon is called **four wave mixing (FWM)**.

Cross-phase modulation (XPM)

Assuming that the conditions are not favorable to FWM, the main contributions to the nonlinear medium polarization are:

$$p_{NL}(t) = \frac{3}{8}\varepsilon_0\chi^{(3)}[(|E_1|^2 + 2|E_2|^2)E_1 e^{j(\omega_1 t - \beta_1 z)} +$$

$$(|E_2|^2 + 2|E_1|^2)E_2 e^{j(\omega_2 t - \beta_2 z)} + \mathrm{c.c.}]$$

For comparison, the nonlinear medium polarization with just one frequency is

$$p_{NL}(t) = \frac{3}{8}\varepsilon_0\chi^{(3)}\left[|E_0|^2 E_0 e^{j(\omega_0 t - \beta z)} + \mathrm{c.c.}\right]$$

Note that the effect of **XPM is twice more intense than SPM**.

The nonlinear refractive index variations at each frequency are

$$\Delta n(\omega_1) \propto n_2 \cdot (|E_1|^2 + 2|E_2|^2)$$

$$\Delta n(\omega_2) \propto n_2 \cdot (|E_2|^2 + 2|E_1|^2)$$

Cross-phase modulation (XPM)

Considering fields like

$$\bar{e}_k(x,y,z,t) = F_k(x,y)\; a_k(z,t) e^{-j\beta_k z}\hat{x} \qquad (k=1,2)$$

and following the same approach used to calculate the NLSE, it can be shown that the complex amplitudes $a_k(z,t)$ obeys the following coupled NLSE

$$\frac{\partial a_1}{\partial z} = -\frac{\alpha}{2} a_1 - \beta_{1,1}\frac{\partial a_1}{\partial t} + j\frac{1}{2}\beta_{1,2}\frac{\partial^2 a_1}{\partial t^2} - j\gamma_1(|a_1|^2 + 2|a_2|^2)a_1$$

$$\frac{\partial a_2}{\partial z} = -\frac{\alpha}{2} a_2 - \beta_{2,1}\frac{\partial a_2}{\partial t} + j\frac{1}{2}\beta_{2,2}\frac{\partial^2 a_2}{\partial t^2} - j\gamma_2(|a_2|^2 + 2|a_1|^2)a_2$$

where the nonlinear coefficients reads $(k=1,2)$

$$\gamma_k \approx \frac{n_2\omega_k}{c_0 A_{eff}}$$

To derive the above equations, we have assumed that the mode field distribution $F_k(x,y)$ do not depend on frequency.

Cross-phase modulation (XPM)

The previous equations confirm that the two waves are coupled through the nonlinear effect. In general, this interaction may lead to complex phenomena, including one similar to modulation instability.

As an example, let us neglect losses and assume that chromatic dispersion at ω_1 is zero. Assume also that $|a_1(z,t)| \ll |a_2(z,t)|$ and wave a_1 is so weak that its SPM can be neglected. The first equation of the coupled NLSE simplifies to

$$\frac{\partial a_1}{\partial z} = -j2\,\gamma_1 |a_2|^2 a_1 \quad \Rightarrow \quad a_1(z,t) = a_1(0,t)\exp\left(-j2\gamma \int_0^z |a_2(z',t)|^2 dz'\right)$$

which shows that, similarly to the case of SPM, in this simplified case the stronger field a_2 induces through XPM a phase modulation on a_1.

This effect is particularly important in **WDM systems**.

Polarization effects

Finally, we consider the case in which the fiber is **anisotropic**, so the light polarization matters. Limiting the analysis to a polarization maintaining fiber, the starting point is the general relation

$$\bar{p}_{NL} = \varepsilon_0 \chi^{(3)} \vdots \bar{e}\bar{e}\bar{e}$$

where the i-th components of the nonlinear medium polarization originates from

$$p_{NL,i}(\bar{e}) = \sum_{j,k,h=1}^{3} \frac{\partial^3 p_i}{\partial e_j \partial e_k \partial e_h} e_j e_k e_h$$

Assume the following field

$$\bar{e} = \frac{1}{2}\left(E_x e^{j\phi_x}\hat{x} + E_y e^{j\phi_y}\hat{y} + c.c.\right)$$

where

$$\phi_{x,y} = \omega t - \beta_{x,y} z$$

(note that the frequency is the same for both polarization components).

Polarization effects

Noting that $e_z = 0$ we have ($i = x, y$)

$$p_{NL,i}(\bar{e}) = \chi^{(3)}_{i,xxx} e_x^3 + 3\chi^{(3)}_{i,xxy} e_x^2 e_y + 3\chi^{(3)}_{i,yyx} e_y^2 e_x + \chi^{(3)}_{i,yyy} e_y^3$$

Performing the explicit products and neglecting phase-mismatched terms, the final result is

$$P_{NL,x} \approx \frac{3\varepsilon_0}{4}\chi^{(3)}\left[\left(|E_x|^2 + \frac{2}{3}\left|E_y\right|^2\right)E_x + \frac{1}{3}\left(E_x^* E_y\right)E_y e^{-j\Delta\beta z}\right]$$

$$P_{NL,y} \approx \frac{3\varepsilon_0}{4}\chi^{(3)}\left[\left(\left|E_y\right|^2 + \frac{2}{3}|E_x|^2\right)E_y + \frac{1}{3}\left(E_y^* E_x\right)E_x e^{j\Delta\beta z}\right]$$

where $\chi^{(3)}$ is the same parameter used for isotropic fibers and $\Delta\beta = \beta_x - \beta_y$. The terms involving $\Delta\beta$ are slightly mismatched, are responsible for FWM phenomena and do not contribute to the nonlinear variations of the refractive indices, which then read

$$\Delta n_x \propto n_2 \cdot \left(|E_x|^2 + \frac{2}{3}\left|E_y\right|^2\right)$$

$$\Delta n_y \propto n_2 \cdot \left(\left|E_y\right|^2 + \frac{2}{3}|E_x|^2\right)$$

Polarization effects

The coupled NLSE results to be

$$\frac{\partial a_x}{\partial z} = -\frac{\alpha}{2} a_x - \beta_{x,1} \frac{\partial a_x}{\partial t} + j \frac{1}{2} \beta_{x,2} \frac{\partial^2 a_x}{\partial t^2} - j\gamma \left(|a_x|^2 + \frac{2}{3} \left| a_y \right|^2 \right) a_x - j \frac{\gamma}{3} a_x^* a_y^2 e^{-j\Delta\beta z}$$

$$\frac{\partial a_y}{\partial z} = -\frac{\alpha}{2} a_y - \beta_{y,1} \frac{\partial a_y}{\partial t} + j \frac{1}{2} \beta_{y,2} \frac{\partial^2 a_y}{\partial t^2} - j\gamma \left(\left| a_y \right|^2 + \frac{2}{3} |a_x|^2 \right) a_y - j \frac{\gamma}{3} a_y^* a_x^2 e^{j\Delta\beta z}$$